

HABITAT FRAGMENTATION AND METAPOPOPULATION, METACOMMUNITY, AND METAECOSYSTEM DYNAMICS IN INTERMITTENT RIVERS AND EPHEMERAL STREAMS

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IN A NUTSHELL

- Intermittent rivers and ephemeral streams (IRES) are dynamic mosaics of lotic (flowing), lentic (nonflowing), and terrestrial habitats
- Dynamic changes in hydrological connectivity shape population, community, and ecosystem patterns and underlying processes in IRES
- Fragmentation can lead to thresholds and tipping points in metacommunity and metapopulation dynamics
- The dynamic changes in connectivity make IRES ideal arenas for exploring temporal dynamics of metapopulations, metacommunities, and metaecosystems from novel perspectives
- Conceptual developments in the dynamics of metapopulations, metacommunities, and metaecosystems have implications for IRES management and conservation (e.g., to define reference conditions and predict restoration trajectories)

4.9.1 INTRODUCTION

Habitat fragmentation is defined as the process during which a large expanse of habitat is transformed into a number of smaller patches of smaller total area isolated from each other by a matrix of habitats unlike the original (Fahrig, 2003). As a result, habitat fragmentation leads to both habitat loss and habitat disintegration, both of which affect biodiversity (Benton et al., 2003; Fahrig, 2003; Haddad et al., 2015). For many species, populations scattered in space are prone to extinction

(Fahrig and Merriam, 1994) if the networks of patches are not sufficiently connected by dispersal (Hanski, 1999; Bowne and Bowers, 2004; Van Dyck and Baguette, 2005). This connection depends on the availability of dispersing individuals and the ease with which these individuals can move across the landscape. This ease of movement is often termed “landscape connectivity” and is a central concept in conservation biology that is of paramount importance for population persistence, patterns of biodiversity, and functioning of ecosystems across landscapes (Fahrig and Merriam, 1994; Kindlmann and Burel, 2008).

Landscape features and geomorphological constraints dictate where and how far individuals can move in all ecosystems thus affecting species coexistence and local community structure (Salomon et al., 2006; Altermatt et al., 2011). In river systems, the dendritic structure of drainage networks results in isolated exterior branches (i.e., the headwaters) converging to form the mainstream channel (Fagan, 2002; Carrara et al., 2012; Chapter 2.1). Moreover, the unidirectional downstream movement of water down river networks carries materials and promotes dispersal of individuals in the downstream direction. Therefore, landscape connectivity and its effects on populations and communities in dendritic networks require a reconsideration of the notion of patch isolation, fragmentation, and connectivity (Fagan, 2002).

Ecological processes operating over multiple spatial and temporal scales determine patterns of population viability and distribution, biodiversity, and ecosystem functioning (Loreau et al., 2003). Specifically, processes structuring populations, communities, and ecosystems occur both at local and large scales, whereby local-scale processes are influenced by processes occurring over much larger scales (Hubbell, 2001; Leibold et al., 2004; Logue et al., 2011). These metasytem concepts (i.e., metapopulations, metacommunities, and metaecosystems) recognize that local discrete populations, communities, and ecosystems form networks connected by gene flow, dispersal, and flows of material and energy. Associated with the recent developments in spatial ecology (Massol et al., 2011), this concept is highly relevant to explore how populations, communities, and ecosystems are organized in intermittent rivers and ephemeral streams (IRES). Stream systems possess several physical characteristics that differentiate them from patch-like systems such as sets of ponds, lakes, or meadows (e.g., Fagan, 2002). For instance, the dendritic structure of drainage networks directs the dispersal of organisms, restricting most dispersal within the stream corridors (Tonkin et al., 2014). Also, the unidirectional flow of water means that passive drift dispersal travels primarily downstream. Such restrictions dictate that dispersal is mostly directional in stream systems, although overland dispersal may also contribute to species distributions across riverscapes. In IRES, the different contributions of spatial and temporal dispersal may complicate assumptions about the importance of directional dispersal for metapopulation and metacommunity dynamics. Therefore, IRES are excellent model systems in which to apply metasytem approaches (Datry et al., 2016a).

In this chapter, we address the effects of riverine network fragmentation by flow intermittence on metapopulation, metacommunity, and metaecosystem dynamics. Considering IRES as coupled aquatic-terrestrial systems, we explore how spatiotemporal patterns of flowing, nonflowing, and dry phases influence the local and regional processes involved in these dynamics. We first present the essential concepts of metasytem and spatial ecology and their relevance to the study of IRES. Then, we review and illustrate metapopulation, metacommunity, and metaecosystem dynamics in IRES, identifying promising research areas. We conclude with a synthesis of research gaps and an illustration of the value of metasytem concepts for the management and restoration of IRES.

4.9.2 CONNECTIVITY, FLUXES, AND DISPERSAL IN RIVER NETWORKS FRAGMENTED BY FLOW INTERMITTENCE

IRES AS DYNAMIC AND SHIFTING HABITAT MOSAICS

IRES are shifting habitat mosaics of lotic (flowing), lentic (nonflowing), and terrestrial habitats whose extent, spatial arrangement, and connectivity constantly vary, following cycles of expansion and contraction (Larned et al., 2010; Datry et al., 2016a; Fig. 4.9.1). Associated with temporal changes in the composition of the mosaic, the longitudinal, lateral, and vertical connectivity between river channels, floodplains, and groundwaters varies greatly (Chapter 2.3). During flow recession, lotic habitats contract and decline in depth and area; lateral aquatic habitats (e.g., floodplains, backwaters, and anabranches) become disconnected from active channels (Boulton, 2003; Chapter 2.3). As contraction continues and flow ceases, the emersion of riffles and other high topographic points along river channels creates longitudinal sequences of disconnected pools (lentic habitats), although these are sometimes connected longitudinally by hyporheic flow (Boulton, 2003). These pools can dry completely due to drainage, seepage, and evaporation, creating terrestrial habitats along dry riverbeds and leading to vertical disconnections with the groundwaters (Chapter 2.3). In some systems, however, disconnected

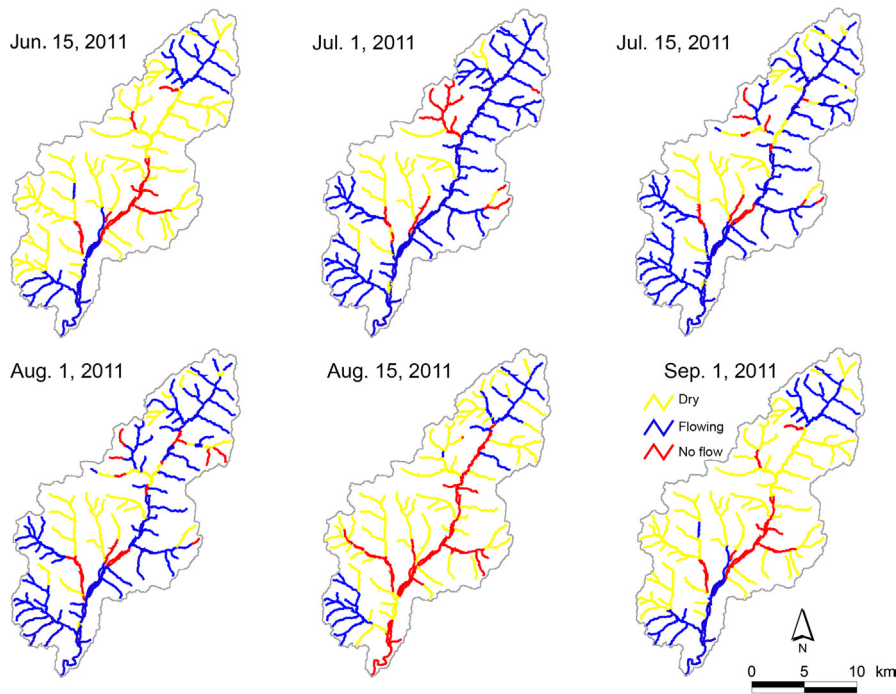


FIG. 4.9.1

Example of a dynamic shifting habitat mosaic of lotic (*blue lines*), lentic (*red lines*), and terrestrial (*yellow lines*) patches in an IRES, the Tude catchment in France during summer, 2011.

pools can persist for months, years, or remain permanent (Hamilton et al., 2005). Flow resumption can take the form of sudden rewetting events when driven by high discharge following rainfall (e.g., Corti and Datry, 2012), instigating a rapid reversal of the drying sequence from terrestrial to lotic habitats. Such events can strongly contribute to groundwater recharge (Chapter 2.3). Rewetting can also be more gradual in large river networks when rainfall is localized to upstream catchments or when driven by rising groundwater levels (Larned et al., 2010). These temporal changes in the mosaic's composition translate into temporal variations in connectivity (Fig. 4.9.1).

FLOW INTERMITTENCE AND CONNECTIVITY

Connectivity can be considered along longitudinal, lateral, vertical, and temporal dimensions in riverine ecosystems (Ward, 1989; Chapter 2.3). Dispersal of organisms and material between habitats occurs in all four dimensions (Fullerton et al., 2010). For example, different stream sites are connected by the longitudinal flow of water. The degree of hydrological connectivity is determined by the presence and amount of water, as well as slope and other geomorphological characteristics. Different sites are also connected by transport of material, such as fine and coarse organic material, usually coming from upstream (Vannote et al., 1980). These materials can be exchanged between terrestrial and stream ecosystems as reciprocal subsidies between streams and riparian zones, thus enhancing connectivity between distinct ecosystems (Soininen et al., 2015) and representing the lateral dimension (Ward, 1989). Stream sites are also connected by the dispersal of organisms, which occurs via hydrological connectivity, active swimming or flight, or passively via aquatic drift or wind (Bilton et al., 2001). These biological dispersal events can be considered as occurring in longitudinal, lateral, or vertical dimensions (Ward, 1989). Vertical connectivity also relates to exchanges of material and organisms between groundwater and surface-water environments. Finally, connectivity typically varies in time, representing the temporal dimension (Ward, 1989).

FLOW INTERMITTENCE AND DISPERSAL

Dispersal is a very important process which affects the distribution of individuals, populations, and species as well as the structure of local communities and metacommunities (Leibold et al., 2004). In IRES, dispersal occurs in both time and in the three spatial dimensions described earlier. *Spatial dispersal* is the movements of individuals between localities and occurs via water, air, or land. In water, organisms crawl or swim actively, drift passively, or move along with other aquatic organisms in the longitudinal dimension (Bilton et al., 2001; Bohonak and Jenkins, 2003; Chapter 4.8). Fish, shrimps, and amphipods are typical examples of active-swimming dispersers in IRES, whereas insect and fish larvae often disperse longitudinally via active or passive drift. Over land, dispersal occurs by flying stages of insects (often adult), passive dispersal by wind, or the use of host organisms as vectors (Bilton et al., 2001, Chapter 4.8). In IRES, an important component of dispersal also incorporates the lateral movements of riparian terrestrial organisms to dry river beds (Corti and Datry, 2016; Chapter 4.4). In particular, these movements may follow diurnal cycles to cope with changes in environmental condition (Corti and Datry, 2016). Additionally, rewetting events can act as mass dispersal events in all three spatial dimensions (Corti and Datry, 2012; Rosado et al., 2015, Chapter 4.3).

The vertical dimension of dispersal of organisms in IRES is probably the least understood but may be highly significant for population and community dynamics (Datry et al., 2016a, Chapter 2.2).

During drying phases, several groups of aquatic organisms are able to actively move into the saturated sediments underneath and lateral to the river channel, defined as the hyporheic zone (e.g., [Vander Vorste et al., 2016a,b](#)). In IRES that maintain connectivity with the hyporheic zone throughout the dry period, the primary source of colonists from the hyporheic zone can greatly outnumber those from longitudinal drift upon rewetting ([Vander Vorste et al., 2016b](#)). Conversely, the hyporheic zone of some IRES can dry up a few hours or days after the surface water recedes in the riverbed (e.g., [Datry et al., 2012](#)) limiting its role in promoting population and community persistence in IRES.

However, the persistence of desiccation-resistant forms in the dry sediments is another way by which populations and communities can be maintained in IRES, and this is termed *temporal dispersal* ([Stubbington and Datry, 2013](#); [Datry et al., 2016a](#)). Temporal dispersal constitutes “traveling in time,” whereby the resting stages of organisms in the sediments can withstand unsuitable conditions and rapidly reappear when conditions become suitable ([Bohonak and Jenkins, 2003](#); [Stubbington and Datry, 2013](#); [Chapters 4.8 and 4.10](#)). In addition, the hyporheic zone can be a refuge during these unsuitable periods and also promote temporal dispersal of aquatic organisms ([Stubbington, 2012](#); [Vander Vorste et al., 2016b](#)).

In the following sections, we explore the effects of the temporal variations of the habitat mosaic composition in terms of lotic, lentic, and terrestrial habitat on biodiversity dynamics and ecosystem processes in IRES. Specifically, we explore how variation in connectivity can influence dynamics at the population, community, and ecosystem scales according to the metasytem perspective.

4.9.3 METAPOPULATION DYNAMICS IN IRES

CONCEPTUAL BACKGROUND OF METAPOPULATION IDEAS

In a given landscape, a set of local populations connected through dispersal and gene flows is referred to as a metapopulation ([Levins, 1969](#)). For organisms that disperse in water, flow cessation and drying of IRES fragments patches of habitat (as water bodies) with little or no movement possible among patches. In IRES, species persistence is governed by the ability of species to survive in isolated water bodies during periods of no flow and colonize newly created habitats during rewatering ([Chapter 2.2](#); [Datry et al., 2016b](#)). The interplay between within-patch persistence and dispersal among patches suggests that the metapopulation concept is an effective framework to address applied and fundamental issues of population and community dynamics in IRES ([Datry et al., 2016a](#)).

[Levins \(1969\)](#) proposed the first modeling framework of the regional persistence of a metapopulation, defined as a set of local populations connected by regional dispersal. [Levins’ \(1969\)](#) model is simple and assumes an infinite number of identical favorable patches within a matrix of inhospitable habitat. Each patch is described by two states: occupied or unoccupied, depending on whether or not the focal species inhabits the patch. An occupied patch may become empty due to population extinction, and an empty patch may become occupied after having been colonized from an occupied patch. [Levins \(1969\)](#) further assumed that colonization rate depends on the proportion of patches occupied. The important message conveyed by his model is that a metapopulation may persist in a landscape only if the colonization rate is higher than extinction rate. If the landscape is modified in a way that extinction rate increases (e.g., patches reduced in size) and/or colonization rate decreases (e.g., more barriers to dispersal), then such changes may drive the entire metapopulation extinct. This reasoning suggests that habitat fragmentation, destruction, or conversion may result in species extinction even if some amount of suitable habitat persists.

In the 1990s, approaches emerged that intended to apply the [Levins \(1969\)](#) framework to real metapopulations within a spatially explicit framework. These approaches also boosted the discipline of metapopulation ecology by providing promising tools to deal with habitat fragmentation, which is a major concern for conservation biologists. For instance, [Hanski \(1994\)](#) introduced the incidence function model (IFM) that could be parameterized simply by fitting a statistical model to data that are widely available (e.g., snapshots of patch occupancies in a landscape). The IFM assumes that extinction rate per patch is an inverse function of its size and that the probability for a patch to be colonized from a second one is a decreasing function of the distance separating them. In contrast with [Levins' \(1969\)](#) initial approach, the IFM is spatially explicit (dispersal occurs between patches with known spatial coordinates), it accounts for heterogeneity among patches in extinction rate through patch size variability, and it deals with a finite number of patches. As in the Levins' model, the regional persistence of a species depends on landscape features. In parallel to these simplifying approaches, more elaborate models have been proposed to generalize the IFM by allowing it to deal with multiple-patch occupancy snapshots of the same landscape taken at different times, leading to more accurate estimates of extinction and extinction rates ([Etienne et al., 2004](#)).

METAPOPULATIONS IN IRES: WHY SO FEW MODELS?

Nevertheless, the potential to apply the framework suggested by [Levins \(1969\)](#) to highly dynamic IRES metapopulations is limited. First, IRES are dendritic structures so that aquatic dispersal is constrained, which contrasts with the two-dimensional dispersal assumed in Levins-type models. This problem obviously matters with strictly aquatic organisms or aquatic phases of amphibiotic organisms, but it may also apply to adult insects with a flight stage that disperse preferentially along the river network (e.g., [Bogan and Boersma, 2012](#)). Experimental studies have shown that, for the same landscape, very different metapopulation dynamics are observed whether or not dendritic dispersal is chosen over a two-dimensional (2D) dispersal ([Carrara et al., 2012](#)). Contrasting metapopulation dynamics are relevant for species such as many aquatic insects that disperse both in the water as larvae and overland as flying adults; the wrong choice of the dominant dispersal mode may lead to the wrong conclusions. When aquatic dispersal is passive (e.g., drift) and dependent on water flow, it is necessarily asymmetric, from upstream to downstream. Asymmetry could also occur in species that disperse actively but that show a preference for upstream (e.g., rainbow trout *Oncorhynchus mykiss*, [Hwan and Carlson, 2015](#)) or downstream movements (e.g., [Davey and Kelly, 2007](#)).

HOW TO DEAL WITH DISPERSAL IN DENDRITIC NETWORKS?

Asymmetric dispersal potentially has consequences for metapopulation persistence (e.g., [Altermatt et al., 2011](#)). Both dendritic and asymmetric dispersal can be dealt with by using spatially explicit Levins-like models such as IFM. The main difficulty arises when several dispersal modes can be used by the same species (e.g., dendritic-aquatic and 2D overland). Ecologists are therefore challenged in accurately estimating the various model parameters when fitting empirical data. Another major difficulty in applying Levins-like models in IRES is the existence of desiccation-resistant stages ([Stubbington and Datry, 2013](#); [Chapter 4.8](#)) and/or the possibility of sheltering in micro-refuges or hyporheic zones ([Vander Vorste et al., 2016a](#); [Chapter 4.8](#)).

The occurrence of resistant life history stages means that a population is not locally extinct even if no functioning individuals are found in a habitat (e.g., during periods of no water), yet the population

can produce dispersing individuals. A third state would need to be added to account for local population presence in cryptic stages and not contributing to dispersal. Additionally, individuals in dormant stages or in “micro-shelters” are generally difficult to sample, which further complicates estimation of extinction and colonization rates in the field. Insights about the main dispersal mode of a species can be obtained by looking at its regional genetic structure (Chaput-Bardy et al., 2008; Chester et al., 2014) or by using manipulative experiments (Vander Vorste et al., 2016b). For instance, Chester et al. (2014) found that some insect species had a genetic signature implying dendritic dispersal whereas others had a signature suggesting overland dispersal within the same river basin.

HOW TO DEAL WITH DYNAMIC AND SHIFTING HABITAT MOSAICS?

Another challenging feature of IRES is their dynamic character in space and time; the number and quality of patches vary in space and time along with dispersal routes connecting them. The first consequence of flow cessation is the gradual loss of lotic habitats from the riverscape, leaving only isolated lentic habitats separated by dry patches (Chapter 2.3). After their isolation, pools or waterholes typically experience changes in their physicochemistry such as increased water temperature and conductivity and decreased dissolved oxygen (Boulton, 2003; Datry, 2017; Chapter 3.1). Additionally, a decrease in the number of patches reduces the number of colonization sources and increases dispersal distances.

Such dynamism of the IRES landscape in terms of quality and number of patches was not covered by the original Levins-like metapopulation model but subsequent models allow for patches that disappear and recur at constant rates within the landscape. A comparison of pulsed dispersal (prior to patch destruction) versus continuous dispersal strategies found that pulsed dispersal increased metapopulation persistence (Reigada et al., 2015). Pulsed dispersal may explain why some species of fishes are more mobile during the onset of the dry period (Matthews and Marsh-Matthews, 2003). Additionally, De Woody et al. (2005) showed that metapopulation persistence is favored when both population extinction and patch destruction probabilities are inversely related to patch size. In IRES, shallow pools and waterholes generally dry up more quickly than large ones, although there are exceptions (e.g., Beesley and Prince, 2010; Chapter 2.2). Interestingly, this model offers great potential for use in real situations because it can deal with spatially explicit landscapes (i.e., dendritic features, directionally biased dispersal). However, there are still too few theoretical works dealing with dynamic landscapes to draw general conclusions or generate hypotheses that could help attain a better theoretical understanding of IRES metapopulations (Datry et al., 2016a).

EXISTING METAPOPULATION STUDIES IN IRES

Modeling real metapopulations in IRES can be extremely challenging if the focal species has stages that (1) can be active or estivating and (2) can disperse through aquatic pathways as larvae and by flying overland as adults (e.g., the caddisfly *Lectrides varians*, Chester et al., 2014). This implies that many parameters should be measured in the field (Lowe, 2002). At the opposite end of the spectrum, the most model-friendly organisms are lentic fishes because they can disperse only via aquatic habitats and few can persist without surface water. Consequently, extinction and colonization rates can be handled realistically with a few parameters (Falke et al., 2012) and the dynamics of the landscape can be readily described by the number, size, location, and longevity of pools or waterholes (Datry et al., 2016b). Moreover, fishes with limited dispersal capacities are also more likely to have difficulties in persisting

as a metapopulation in IRES than most invertebrates. For this reason, fishes frequently gain attention from conservation biologists (Labbe and Fausch, 2000; Jaeger et al., 2014), and the few quantitative studies that have addressed metapopulation issues in IRES focus on fishes.

The first step in conducting a quantitative metapopulation study in IRES is to estimate extinction and colonization rates, while accounting for patch creation (e.g., rewetting) and destruction (e.g., pool drying) if the landscape is dynamic. Such approaches are still rare (e.g., Falke et al., 2012) but are expected to be useful in comparing metapopulation dynamics among species, among years, or in the same study among streams differing in their level of flow intermittence.

Quantifying both landscape and occupancy dynamics is also the first step toward more comprehensive modeling in which the persistence of a metapopulation is explicitly described to quantify the impact of different natural or anthropogenic environmental changes. Few studies have incorporated these latter features. In an IRES in the Murray-Darling Basin in Australia, a spatially explicit metapopulation model for golden perch (*Macquaria ambigua*), present in isolated waterholes during the dry season, estimated metapopulation persistence under different scenarios including climate changes and human water extraction (Bond et al., 2015). One interesting result was the high modeled vulnerability of the metapopulation when surface water extraction preferentially targets large waterholes because this mirrors expectations from the simpler and more general model of De Woody et al. (2005). According to this model, metapopulation persistence is lowered when the probability of patch destruction (e.g., waterhole drying due to surface water extraction) is correlated with the size of the patch (e.g., large waterholes). Conversely, the metapopulation may cope better with a scenario of increasing aridity because, in this case, the probability of drying is inversely related to waterhole size. This study illustrates how different landscape dynamics may have contrasting consequences on metapopulation persistence. Both population and landscape dynamics need to be accounted for to provide insightful guidelines for metapopulation management.

Many studies solely consider landscape changes under different scenarios to assess impacts on a few species of interest, such as mapping aquatic refuges and their isolation under different hydrological regimes. Modeling the dynamics of the landscape is a necessary but not sufficient step and is improved by better integration of climatic, hydrological, and human extraction models (e.g., Falke et al., 2012); the availability of physical-chemical sensors allowing for data logging (e.g., Larned et al., 2011); and the use of citizen scientists in river monitoring to describe dynamics of habitat mosaics in IRES (e.g., Datry et al., 2016b).

WHERE TO GO FROM HERE?

Metapopulation theory suggests that a species will become extinct in a landscape before the suitable habitat (e.g., refuges) is entirely gone. Consequently, the next step is to build on our increasing ability to model landscapes and riverscapes in space and time to add true metapopulation dynamics and to accept “the ephemeral nature of local populations while planning and monitoring for regional persistence” (Labbe and Fausch, 2000). As demonstrated by Falke et al. (2012) and Bond et al. (2015), landscape and occupancy dynamics could and should be estimated jointly to derive a reliable understanding of the fate of a metapopulation and to assess its long-term persistence when faced with different management options. However, this is mainly true for fishes. For species with more complex life cycles combining active and dormant stages and/or different dispersal modes, the parameterization of a metapopulation model is so challenging that it could prevent empirical field-based approaches. The relevance of metapopulation

ecology to management of environmental scenarios has been improved by the development of general and simple models that are easily parameterized with few field data. Such a framework is largely lacking for metapopulations embedded within a dynamic landscape, such as IRES. This stresses the need for more empirical and theoretical works to identify which questions could be answered using general and simple models and which ones require more specific and data-demanding models.

4.9.4 METACOMMUNITY DYNAMICS IN IRES

CONCEPTUAL BACKGROUND OF METACOMMUNITY IDEAS

Metacommunity ecology examines how local dynamics of ecological communities are associated with regional dispersal of species (Leibold et al., 2004). Consistent with metapopulation theory, dispersal is an integral part of the four main metacommunity perspectives. First, in species sorting, sufficient dispersal is necessary so that species are able to track variation in abiotic and biotic environmental conditions among sites. Hence, with sufficient dispersal, all species in the regional species pool (i.e., a metacommunity) can potentially reach all sites, and a subset of species is filtered to occur at a site according to its environmental conditions. Second, with mass effects, high dispersal rates homogenize, at least to some extent, the community structure of adjacent sites even though they may possess different environmental conditions. Mass effects through high dispersal rates may also decouple local communities from purely environmental control.

Third, in neutral dynamics, random speciation, extinction, immigration, and emigration processes structure communities of functionally equivalent species (Hubbell, 2001). Community structure should be different between distant sites, with community similarity decaying along spatial gradients irrespective of any environmental influences. Fourth, in patch dynamics, a colonization-competition trade-off dictates that some species are good dispersers but poor competitors, whereas other species are poor dispersers but good competitors. Again, dispersal limitation plays an important role in patch dynamics. While these four perspectives are useful in metacommunity studies, they are rarely mutually exclusive and many, if not most, natural freshwater metacommunities show signs of multiple perspectives (Heino et al., 2015a,b).

HOW TO STUDY METACOMMUNITIES IN IRES?

Metacommunity organization has been studied using both observational and manipulative approaches. Observational approaches have analyzed field data on species distributions using multivariate analytical methods to infer the relative roles of different assembly processes. Commonly used multivariate analytical approaches include using spatial eigenfunction analysis in association with constrained ordination to tease apart environmental and spatial (e.g., dispersal) effects. Another approach has been to use distance-based methods, such as Mantel tests and regression of distance matrices to associate biological and environmental distance matrices (distance-decay relationships, Legendre and Legendre, 2012). While the advantages and disadvantages of these two descriptive approaches have been discussed intensively (Legendre et al., 2005; Tuomisto and Ruokolainen, 2006), they may be seen as complementary. The ordination-based approach may be better in describing community-environment relationships, whereas the distance-based approach may be better in examining true effects of distance on the decay (i.e., change) of community similarity along geographical gradients.

Experimental approaches include the use of meso- and microcosms (Altermatt et al., 2013), where local environmental conditions, species interactions, and dispersal rates can be manipulated. Although experimental approaches are highly efficient when focusing on the mechanisms affecting metacommunity organization, they may only be conducted at unrealistic scales that are too small to validly extrapolate to natural situations. On the other hand, observational field-based studies are plagued by difficulties in associating patterns with mechanisms, typically because of missing abiotic and biotic variables as well as spatially autocorrelated environmental variables (Jacobson and Peres-Neto, 2010; Heino et al., 2015a,b). The latter problem largely precludes meaningful comparisons of environmental and spatial effects on local community structure in field-based studies.

SPECIES TURNOVER AND NESTEDNESS IN IRES METACOMMUNITIES

The terms “nestedness” and “turnover” are directly related to interpreting and explaining variation in community composition among sites (i.e., beta diversity, Anderson et al., 2011) and are strongly related to metacommunity organization. Variation in community composition among multiple locations varies due to the replacement of species among sites (e.g., sites support unique species), species loss (e.g., one site supports a subset of species present at another), or a combination of both processes (Baselga, 2010; Legendre, 2014; Fig. 4.9.2). These concepts are useful for the development of basic theory of biodiversity by illustrating how local communities are assembled, and also for applied conservation to identify which communities (locations) support most of regional biodiversity.

Increasing duration or prevalence of flow cessation can filter species out and cause local communities from IRES to be nested within perennial ones (Larned et al., 2010; Chapter 4.8). This may explain why in many situations alpha diversity of aquatic species declines with increasing duration of flow cessation (e.g., Datry et al., 2014a; Table 4.9.1). Habitat fragmentation is also linked to nestedness,

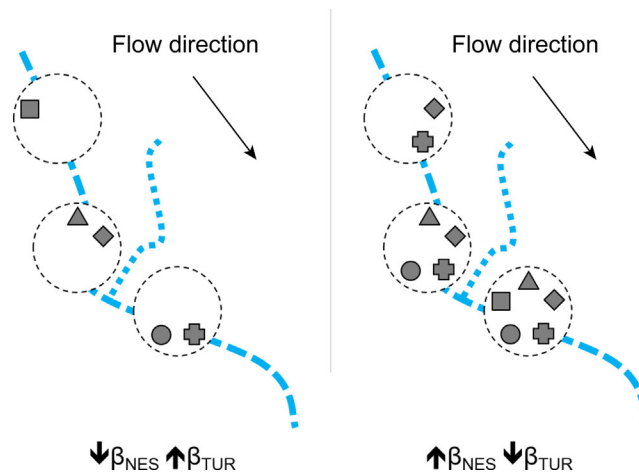


FIG. 4.9.2

Turnover (β_{TUR}) refers to the replacement of species among samples, whereas nestedness (β_{RIC}) indicates the degree to which species-poor communities are a subset of relatively richer communities. Different-shaped symbols represent the different species in each local community distributed along the intermittent river network (dashed circle).

Table 4.9.1 General effects of flow intermittence on biodiversity in IRES across different spatial components of species diversity

Component of diversity ^a	Potential response to flow intermittence	Organism group	Examples
Alpha diversity (number of species present in a location)	Declines with increasing duration or frequency of drying	Riparian and floodplain plants	Stromberg et al. (2005)
	Unimodal peak in alpha diversity at intermittently inundated locations compared to permanently or ephemerally inundated sites	Riparian plants	Katz et al. (2012)
	Declines with decreased habitat persistence	Fish, macroinvertebrates	Datry et al. (2007) , Beesley and Prince (2010)
	Declines with increasing habitat fragmentation and isolation	Bacteria, plankton, macroinvertebrates	Fazi et al. (2013)
	Negatively associated with habitat size (minimum discharge)	Macroinvertebrates, fish	McHugh et al. (2015)
Beta diversity (spatial variation in species composition among locations)	Change in community composition along a gradient of hydrological intermittence (drying, discharge permanence)	Riparian plants, fish, macroinvertebrates, algae	Datry et al. (2014a)
	Communities become more spatially homogenous during periods of low discharge and during and immediately after flooding, and show greatest variation when hydrological conditions are between extremes of flooding and drying	Macroinvertebrates, fish	Fernandes et al. (2009) , and Buendia et al. (2014)
	Spatial variation in composition along intermittence gradients is driven by loss (nestedness) or turnover (replacement) of species due to increasing fragmentation or environmental harshness	Macroinvertebrates, fish	Bogan et al. (2013) , Miyazono and Taylor (2015) , Datry et al. (2014a, 2016a) , Ruhí et al. (2015)
Gamma diversity (number of species present in a region)	Less well studied in IRES; however, evidence indicates that intermittent river reaches contribute to gamma diversity for an entire river network	Macroinvertebrates	Larned et al. (2010)

^aDefinitions based on terminology of [Whittaker \(1960\)](#) and [Anderson et al. \(2011\)](#).

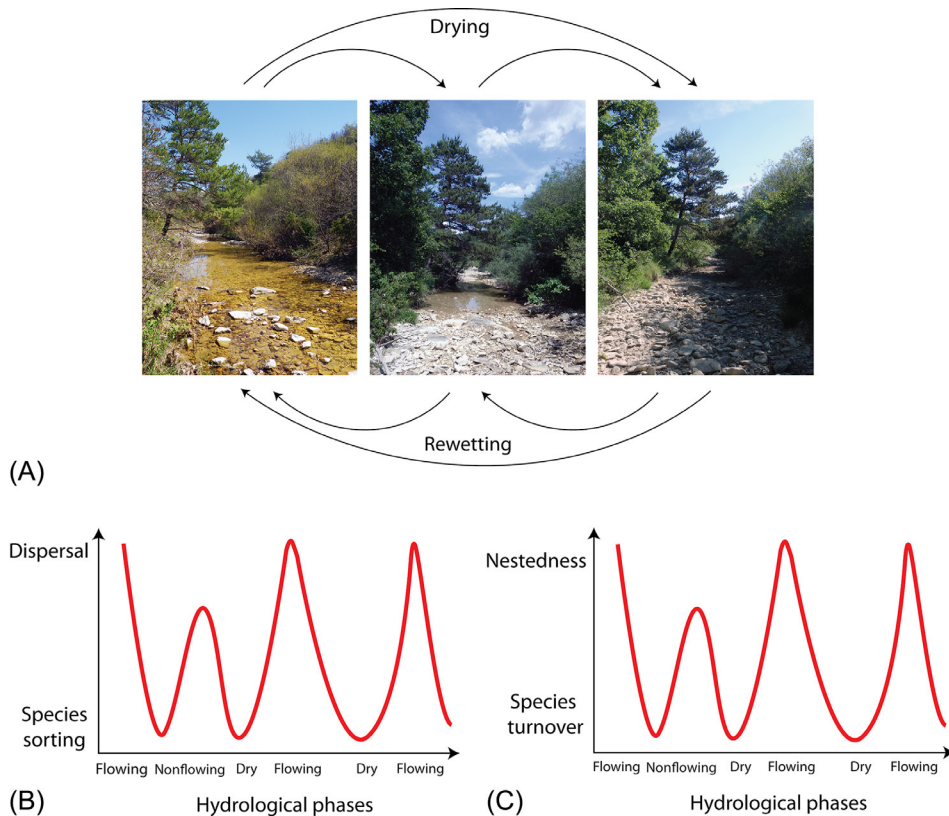
whereby communities in sites that experience prolonged fragmentation are often highly nested subsets of connected habitats (Miyazono and Taylor, 2015; Datry et al., 2014a). In some rare cases, compositional differences between intermittent and perennial sites can be driven by species turnover, where communities in intermittent sites support taxa that are specialized to occur and dominate in such sites (Chapters 4.1–4.5). Turnover among contrasting flow regimes has been observed for some invertebrate and diatom communities (Bogan et al., 2013; Tornés and Ruhí, 2013). As the prevalence of turnover and nestedness varies among organism groups and regions, it may be hypothesized that the degree to which they relate to spatial variation in composition depends on a combination of dispersal mode, regional species pool selection, and long-term hydrological history. However, this hypothesis requires further development and testing (Tonkin et al., 2015; Datry et al., 2016c; Leigh and Datry, 2017).

METACOMMUNITIES IN IRES

The alternation of contrasting hydrological phases in IRES generates dynamic variation in the respective metacommunity assembly processes, which may lead to dynamic community patterns in the network (Cañedo-Argüelles et al., 2015; Datry et al., 2016b,c). For example, shifts from flowing to nonflowing conditions are likely to cause a rapid increase in the importance of species sorting, including adaptations to environmental conditions that occur during flow cessation, biotic interactions within contracting pools, and predation by terrestrial organisms (Fig. 4.9.3). Subsequently, the relative importance of dispersal in accounting for community structure increases with the arrival of large specialist predators with strong flying abilities, such as dragonflies (Odonata), diving beetles (Coleoptera), and some true bugs (Heteroptera) (Bogan and Boersma, 2012; Bonada et al., 2006; Chapter 4.8; Fig. 4.9.3). Rewetting of habitats can promote colonization via two processes: dispersal of organisms from refuges (Datry et al., 2014a; Chapter 4.8) and emergence of desiccation-resistant forms from rewetted sediments (temporal dispersal, Stubbington and Datry, 2013; Chapter 4.8).

The relative importance of these two recolonization processes likely depends on two factors. The first is the duration of substrate drying, where viability of desiccation-resistant forms can decrease with increasing drying duration. The second is the geographic distance from refuges supporting source of colonists, where recolonization via dispersal is more rapid when refuges are close compared to locations separated by long distances. After these first phases of recolonization involving dispersal, species sorting becomes gradually more important (Fig. 4.9.3). Similar temporal dynamics can be predicted for semiaquatic and terrestrial communities in IRES, although the restructuring processes involved during each phase have been poorly explored thus far.

Consequently, the structure of a metacommunity may reflect the temporal variability in the relative roles of community assembly processes. The most obvious example is probably the respective proportions of lotic, lentic, and terrestrial species at a given location. Community structure may vary sharply during the different hydrological phases, with notable dominance by lotic species during flowing phases, by lentic species during nonflowing phases, and by terrestrial species during dry phases (Bonada et al., 2006; Corti and Datry, 2016). Many metrics of community structure, such as alpha diversity, species relative abundances, or the proportion of predatory species may undergo abrupt changes that include “step changes” in response to changes in hydrological conditions (Boulton, 2003). The temporal variability in community assembly processes may also lead to predictable spatial patterns of metacommunity organization. During phases dominated by dispersal, nestedness may be observed, particularly for weak to moderate dispersers (Larned et al., 2010; Datry

**FIG. 4.9.3**

Alternating cycles of flowing, nonflowing, and dry phases in IRES (A) provide the dynamic environment for two hypothetical examples of how these hydrological phases and shifts influence the respective importance of species sorting and dispersal on metacommunity organization (B) and potential changes in spatial organization of communities (C). Panels B and C are derived from [Datry et al. \(2016a\)](#).

Photos: Courtesy B. Launay.

[et al., 2014a; Datry et al., 2016c; Fig. 4.9.3](#)). In contrast, when species sorting dominates, species turnover may be observed more commonly ([Datry et al., 2016b; Tonkin et al., 2015; Fig. 4.9.3](#)).

Metacommunity patterns in IRES are likely to alternate in response to alternating flowing, nonflowing, and dry phases. Consequently, snapshot spatial views of metacommunities are likely to provide incomplete understanding of which community assembly processes are structuring local community composition. Although the temporal dynamics of metacommunities have not been explored in IRES to date, evidence from other ecosystems supports these predictions. For example, recurrent hurricanes temporarily reduce the nestedness of gastropod assemblages in tropical wet forests ([Bloch et al., 2007](#)). In seasonal floodplains, fish metacommunity structure varies between the initial and late phases of the flooding in response to a shift in the importance of connectivity (cf. dispersal) versus local environmental conditions (cf. species sorting) in structuring local communities ([Fernandes et al., 2009](#)). In perennial river

systems, different models of metacommunity structure apply along gradients of flooding disturbance (Campbell et al., 2015) and environmental harshness (Datry et al., 2016c). In clusters of constructed ponds, environmental filtering by drought increased nested structures in invertebrate metacommunities (Ruhí et al., 2013). Finally, in temporary wetlands, the transitions from terrestrial to aquatic phases modify community assembly processes, and the importance of species sorting decreases with inundation time (Vanschoenwinkel et al., 2009).

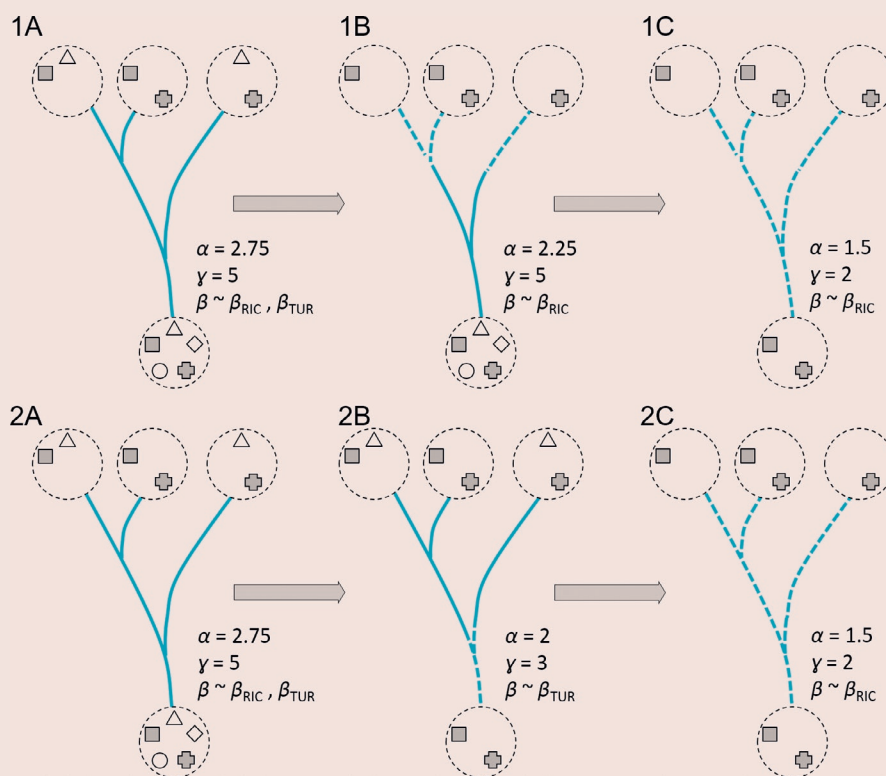
Compared to other temporary systems, such as rock pools, vernal pools, wetlands, or intertidal zones, the temporal variability in the relative roles of community assembly processes associated with flow cessation and drying may differ spatially in IRES due to the dendritic nature of river networks and directionally biased dispersal (Brown and Swan, 2010; Altermatt et al., 2011; Datry et al., 2016a). Contrary to the paradigm that headwater communities are driven by species sorting whereas those of lowland downstream reaches are driven by dispersal due to convergence of all channels and downstream water flow (Brown and Swan, 2010), more complex patterns probably occur in IRES. This is because of distinctive configurations of flow cessation and channel drying. For example, headwater communities of a network whose drying reaches are upstream may be shaped more by dispersal limitation than by species sorting, since the sources of colonists are downstream or in either the saturated sediments or dry underlying sediments (Chapter 4.8; Datry et al., 2016a). In turn, metacommunities of IRES are likely to show patterns that vary spatially depending on the location of drying events in the network. For example, nestedness related to dispersal may predominate in mid-reaches or downstream drying systems (e.g., Datry et al., 2014a,b), whereas turnover related to species sorting should be more common in systems with intermittent headwaters or drying out completely (e.g., Fagan, 2002). Although these ideas remain speculative, the emergence of metacommunity studies in IRES allows for their further testing.

BIODIVERSITY PARTITIONING IN IRES

Biodiversity has multiple spatial components (Whittaker, 1960), and different components of biodiversity vary along hydrological gradients in IRES (Table 4.9.1; Box 4.9.1). Local species richness (i.e., number of taxa within a location) is termed alpha (α) diversity, the overall number of species in a region

BOX 4.9.1 PARTITIONING OF SPECIES DIVERSITY IN IRES

The combination of flow directionality, associated drift, and network topology means that an upstream local assemblage may share more species with a downstream local assemblage than with a local community of a different headwater site (Fig. 4.9.4). This is why, in the absence of drying, the turnover component of beta diversity may largely explain differences in composition among headwaters, and differences in the richness component are more important between headwater and downstream (main stem) communities. Under this scenario, both components contribute to overall beta diversity in the metacommunity (1A and 2A in Fig. 4.9.4). The unevenness in the distribution of dissimilarity across the dendritic network is because if drying starts in the headwaters (1B in Fig. 4.9.4), the metacommunity may be homogenized via an increase in the nestedness component (all headwaters lose idiosyncratic, tolerant species). In contrast, mainstem drying (2B in Fig. 4.9.4) may reduce the relative importance of this component and increase that of turnover (in 2B, the mainstem has lost tolerant species that were contributing to nestedness in 1B). Finally, when low- or nonflowing conditions affect the whole river network (1C and 2C in Fig. 4.9.4), only tolerant species persist. At this point, dissimilarity across local communities is determined by the tolerant species that constituted the local communities before drought (Ruhí et al., 2013). Complementary patterns in alpha, beta, and gamma diversities of terrestrial faunas colonizing the dry river beds could be expected but have not yet been described.

BOX 4.9.1 PARTITIONING OF SPECIES DIVERSITY IN IRES (Cont'd)**FIG. 4.9.4**

Predicted patterns of alpha diversity (α), beta diversity (β), and its components (richness: β_{RIC} turnover: β_{TUR}), and gamma diversity (γ) overtime. The first row represents drying (*dashed blue line*) starting in the headwaters; the second row represents drying starting in the main stem. Dashed circles represent local communities containing species (various symbols) that are either tolerant (shaded symbols) or sensitive (unshaded symbols) to low-flow or no-flow conditions.

is termed gamma diversity (γ), and variation in the composition of local communities within regional units (i.e., across locations) or within a location or groups of locations (i.e., over time) is termed beta (β) diversity (Anderson et al., 2011). Compared to both alpha and gamma diversities, beta diversity is conceptually more complex because there are multiple aspects of communities that can be used to describe variation among locations. For example, variation in beta diversity can be explained by variation in richness differences, in the identity of the species, or in their relative abundances (see Anderson et al., 2011; Box 4.9.1). Accordingly, this topic has attracted much interest from both fundamental and applied perspectives (Baselga, 2010; Legendre and Cáceres, 2013; Legendre, 2014). Importantly, each component of diversity is relevant to aspects of spatial scaling (e.g., grain size and spatial extent) that is

often determined by researchers on the basis of the basic biology of study taxa, scale of environmental variables of interest to patterns of diversity, and geographic variables (e.g., drainage divide, biogeographical pool).

WHERE TO GO FROM HERE?

Metacommunity research in IRES is progressing rapidly but the temporal dynamics of metacommunities in IRES and other dynamic systems remain unexplored. Whereas dispersal processes may be of paramount relevance during high-flow phases, species sorting may be more important during nonflowing or dry phases. This temporal variability is certainly enhanced by discrete and punctuated biotic interactions between aquatic and terrestrial communities in IRES and other coupled aquatic-terrestrial systems. In the case of dendritic systems with directionally biased dispersal, the location and spatial extent of disturbances, such as drying, may interact with the temporal variations in community assembly processes to produce complex spatiotemporal variability in local community structure. In the context of increasing extreme climatic events and ecosystem disturbances, understanding how metacommunities are organized in highly dynamic systems is becoming a key research topic. Translating this research into efficient management guidelines is urgently needed (Section 4.9.6).

4.9.5 METAECOSYSTEM DYNAMICS IN IRES

CONCEPTUAL BACKGROUND OF METAECOSYSTEM IDEAS

As a natural extension of metapopulation and metacommunity theory, metaecosystems represent a set of local ecosystems connected by spatial flows of energy, materials, and organisms across ecosystem boundaries (Loreau et al., 2003; Table 4.9.1). The metaecosystem concept recognizes that ecosystems act as sinks and sources of subsidies and organisms, such that different ecosystems fulfill different functions (Loreau et al., 2003). A striking example is the lateral reciprocal subsidies between streams and riparian zones (Polis et al., 1997). The concept was also extended to dendritic ecological networks such as river networks across which carbon is being continuously supplied, processed, and stored in habitats ranging from headwaters, river floodplains, and the hyporheic zone to terminal lakes, estuaries, and oceans (Vannote et al., 1980; Battin et al., 2009).

IRES AS METAECOSYSTEMS: INSIGHTS FROM ORGANIC MATTER DYNAMICS

IRES are ideal systems to develop further the metaecosystem perspective of river networks owing to their dynamic mosaic of flowing, nonflowing, and dry habitats (Section 4.9.2). We contend that the dynamic mosaic functions as a set of local ecosystems contributing differentially to the transport, deposition, and processing of material. In the absence of empirical synthesis studies, we illustrate this metaecosystem perspective of IRES using examples derived from field studies on coarse particulate organic matter (CPOM), but the same insights are also applicable to dissolved organic carbon, nutrients, and autotrophic carbon.

In comparison to flowing habitats, the longitudinal and lateral transport of CPOM is halted in nonflowing and dry habitats (Boulton, 1991; Langhans et al., 2008; Corti et al., 2011) and CPOM

**FIG. 4.9.5**

Accumulation of CPOM in the dry channels (a, b) and disconnected pools (c, d) of IRES. Terrestrial vegetation encroaches onto the alluvial sediment when surface water dwindles (e, f). The sites are as follows: Demnitzer River, Germany (a), Albarine River, France (b, c, e, and f), and the Eygue River, France (c).

Photos: Courtesy R. Corti.

from riparian zones accumulates and vegetation develops on dry sediments (Boulton, 1991; Corti et al., 2011; Fig. 4.9.5). The types of processes involved in CPOM decomposition and their primary agents also differ within the dynamic mosaic (Chapters 3.1 and 3.2). In flowing and nonflowing habitats, decomposition rate decreases with the duration of past and current drying events, due to the loss of aquatic fungi and shredders (Bruder et al., 2011; Corti et al., 2011; Datry et al., 2011).

In dry habitats, the decomposition rate is reduced to almost zero due to the low abundance of terrestrial shredders, reduced microbial activity, absence of mechanical breakdown, and low leaching rates (Maamri et al., 2001; Corti et al., 2011). However, both dry and nonflowing periods can induce substantial variability in organic matter chemistry, bioavailability, and decomposition rates (Dieter et al., 2013; Chapter 3.2), due to the effects of leaching during rain events, photodegradation by ultraviolet light (Austin and Vivanco, 2006; Fellman et al., 2013), fermentation and accumulation of toxic compounds in anoxic pools (Dieter et al., 2013; Canhoto et al., 2013), and nutrient uptake by microbes and invertebrates in standing pools and in the hyporheic zone (Corti et al., 2011; Febria et al., 2012).

Alternation of flowing, nonflowing, and dry phases controls the flow of material within the habitat mosaic and connects the set of ecosystems (Larned et al., 2010; Datry et al., 2014b). Parallel to the predictions of the River Continuum Concept (Vannote et al., 1980), CPOM in IRES networks is progressively transported to be stored and processed in retention sites downstream; the river network functions as a longitudinal biogeochemical reactor (Battin et al., 2009, Fig. 4.9.6). However, unlike perennial networks, IRES function as a punctuated longitudinal biogeochemical reactor (Larned et al., 2010). While flow is absent, transport of CPOM among patches is inhibited and ecosystems are isolated (Fig. 4.9.6). When flow resumes, CPOM is transported and stored downstream, especially when floodwaters recede (Corti and Datry, 2012; Rosado et al., 2015). The quantity of transported CPOM can be highly variable, depending on climate and seasonal variations in leaf litter inputs (Corti and Datry, 2012). Consequently, the frequency, timing, and magnitude of rewetting events control the degree of connectivity among ecosystems; frequent, long, and intensive rewetting events increase the exchange of CPOM among habitats.

The spatial organization of the habitat mosaic controls the transport, storage, and processing of CPOM, ultimately influencing its quantity, quality, and bioavailability downstream (Box 4.9.2). Downstream waters are expected to receive more material when dry habitats are located upstream, compared to perennial networks and networks with only downstream dry habitats. This is because of the combined effect of lower rates of decomposition and increased accumulation in dry patches (Box 4.9.2). In the temperate Albarine River in France, for instance, the concentration of transported CPOM at the leading edge of a rewetting front advancing over a dry section exceeded 600 g m^{-3} in autumn, and was 50-fold higher than the concentration transported in the upstream perennial section (Corti and Datry, 2012). Large, sudden pulses of CPOM transport and storage can produce “hot moments” of biogeochemical transfer and transformation (Datry et al., 2014b), reducing the quantity of processed material received downstream. However, the processing of the massive input of terrestrial labile compounds can also cause dissolved oxygen to plummet, killing invertebrates and fishes (Hladysz et al., 2011; Chapter 3.2) and exporting unprocessed material. The bioavailability of the CPOM reaching downstream habitats also depends on chemical transformations occurring upstream or on floodplain habitats. For instance, the decomposition of leaves previously affected by solar radiation in upstream dry habitats can be reduced by 33% (Dieter et al., 2013). In addition, the mixing at confluences of CPOM previously processed in different habitats could generate either positive or negative effects on decomposition (Gartner and Cardon, 2004; Hättenschwiler and Gasser, 2005). Ultimately, the effects of the different processes involved in the processing, storage, and transport of CPOM along IRES are likely to propagate down the river network and influence the annual CPOM budget.

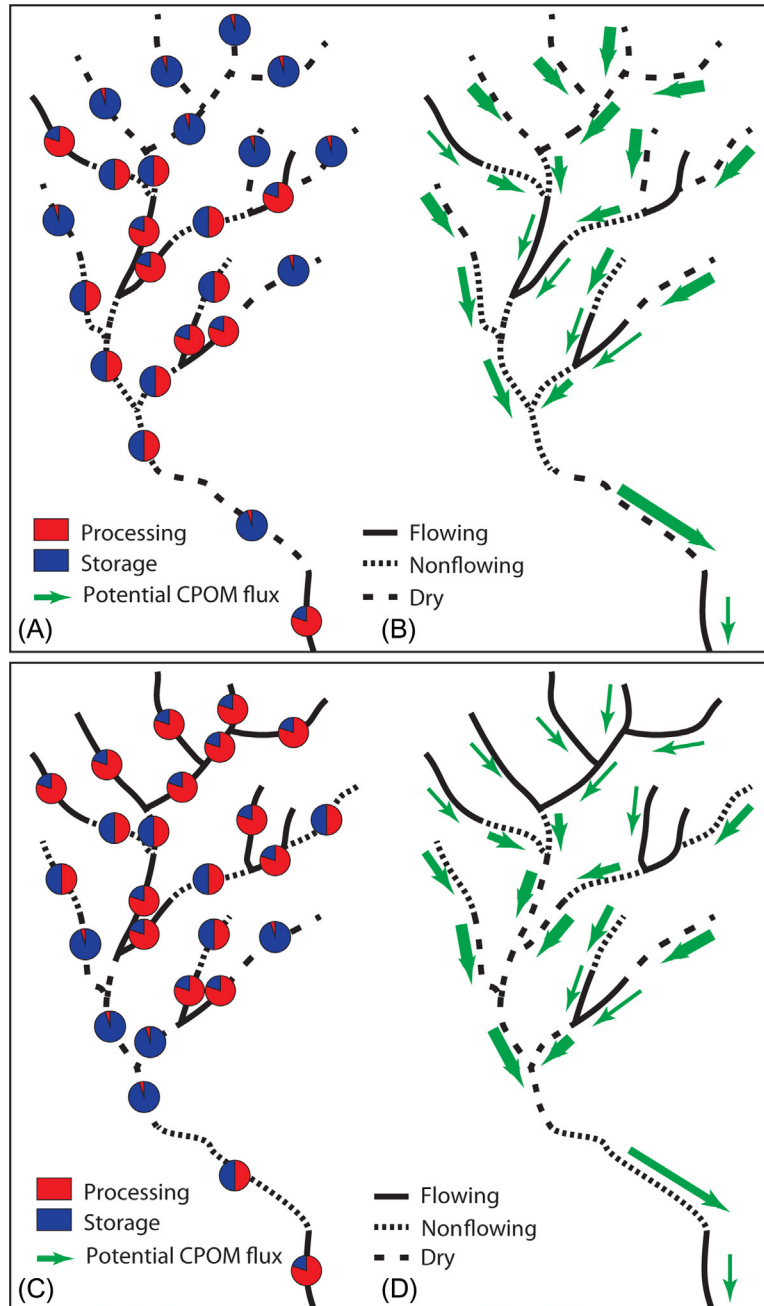


FIG. 4.9.6

Two theoretical networks of IRES (A, C) composed of flowing, nonflowing, and dry habitats. The contribution by each habitat to the processing and storage of CPOM differs among habitats; flowing habitats have higher decomposition rates whereas dry habitats have higher accumulation rates (B, D). The potential contribution of habitats to CPOM fluxes also varies with habitat and is greatest in dry habitats. In addition, the spatial organization of flowing, nonflowing, and dry habitats across the network determines the potential fluxes of CPOM among the habitats of the mosaic (see [Box 4.9.2](#)).

BOX 4.9.2 COARSE PARTICULATE ORGANIC MATTER DYNAMICS IN IRES WITH CONTRASTING SPATIAL PATTERNS OF FLOW INTERMITTENCE

Integrating the roles and contributions of IRES to the dynamics of coarse particulate organic matter (CPOM) at the scale of entire river networks (from headwaters to oceans or terminal lakes) could greatly alter our current perception of the role of rivers in the global carbon cycle (Larned et al., 2010; Datry et al., 2014b). We contend that current global estimates of carbon (C) fluxes might be inaccurate because the role of IRES in the storage, processing, and release of organic material has not yet been considered.

The spatial arrangement of flowing, nonflowing, and dry habitats within IRES networks is likely to induce pronounced differences in C fluxes compared to networks entirely composed of perennially flow habitats (perennial-river network). In a perennial-river network (Fig. 4.9.7A), CPOM is progressively stored and processed during its transport downstream and, with distance downstream, CPOM quantity increases due to the increase in retention structures. In a river network with upstream IRES (Fig. 4.9.7B), dry and nonflowing habitats temporarily reduce the quantity of CPOM delivered

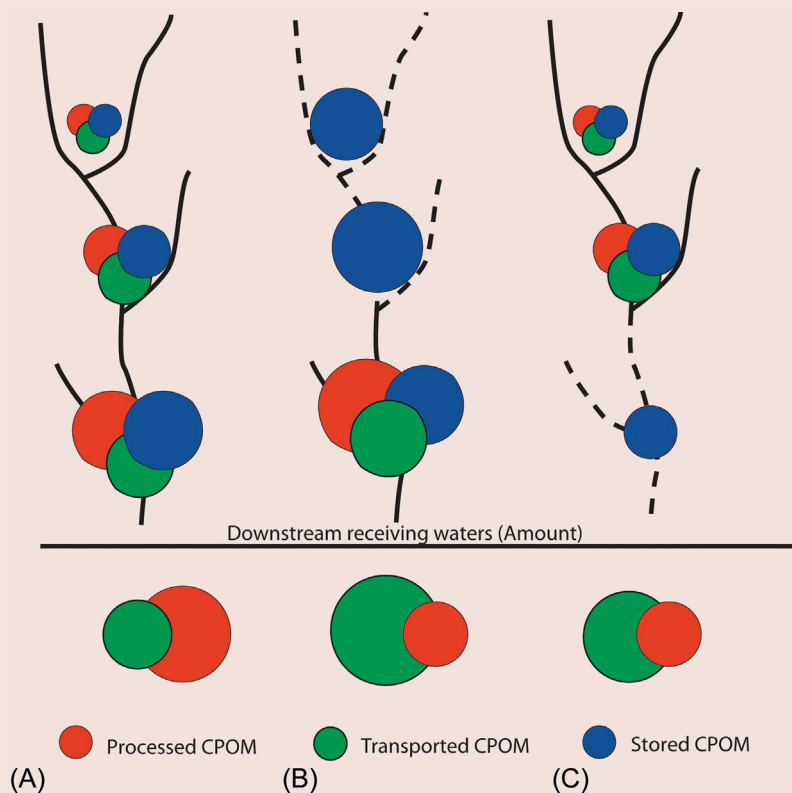


FIG. 4.9.7

Schematic representation of three geomorphically identical river networks with different spatial flow patterns (A-C) showing potential variation in the quantity of processed (*red circles*), transported (*green circles*), and stored (*blue circles*) CPOM, and the annual budget of CPOM in downstream receiving waters. Circle diameter is proportional to the quantity of CPOM; dashed lines represent intermittent reaches and solid lines represent perennial reaches.

BOX 4.9.2 COARSE PARTICULATE ORGANIC MATTER DYNAMICS IN IRES WITH CONTRASTING SPATIAL PATTERNS OF FLOW INTERMITTENCE (Cont'd)

downstream and potentially decrease how much CPOM is processed within the network. Consequently, the annual amount of processed CPOM in downstream receiving waters is expected to be lower compared to a perennial-river network whereas the annual amount of unprocessed transported CPOM might be greater, due to sudden huge pulses of CPOM carried during rewetting events coupled with low decomposition rates in intermittent habitats.

In a river network with downstream IRES, smaller quantities of CPOM are expected to accumulate in dry habitats due to lower inputs of CPOM from riparian zones in larger streams. Consequently, the amount of material reaching downstream waters is greater than in perennial-river networks but lower than in networks with upstream IRES (Fig. 4.9.7C). To determine how biogeochemical differences between IRES and perennial-river networks might challenge current estimates of C fluxes in river networks, large-scale measurements of CPOM deposited along riverbeds, their changes over time, and estimates of the biodegradability of the organic matter are needed. These measurements must be supplemented with concurrent data on the environmental drivers controlling these processes (e.g., riparian cover, climate, substrates type, drying duration, and frequency) and will yield crucial insights into global dynamics of C and their likely changes in response to climate change and other factors affecting flow regimes of rivers worldwide.

WHERE TO GO FROM HERE?

Increasing knowledge about transport, accumulation, and processing rates during the different hydrological phases combined with spatial and metamodeling approaches is a promising avenue to model CPOM dynamics in IRES. Incorporating IRES into global carbon models and flux estimates is clearly important but will be challenging. Metasystem approaches can also be used to include adjacent systems (e.g., riparian zones, groundwaters) when exploring biodiversity and ecosystem processes in IRES.

4.9.6 RESEARCH AND MANAGEMENT PERSPECTIVES

One theme in this chapter is that the metasystem approach opens up numerous perspectives for both basic and applied IRES ecology. Although there are many questions to explore, we list some of the main points and challenges that we believe should be addressed using metasystem approaches to improve our understanding of the processes structuring ecosystems across spatial scales in IRES.

Metapopulation modeling in IRES is challenging, yet adapting approaches developed in more static systems would increase our ability to predict species distributions in these and other dynamic ecological systems. Increased ability to predict metapopulation persistence in IRES would be highly valuable for species conservation issues in the face of global changes. For example, accounting for desiccation-resistant stages in metapopulation models will represent an important advance.

Metacommunity organization in IRES should continue to examine the link between community variation across space and underlying mechanisms driving differences in community composition. One promising approach is to incorporate spatial and temporal variation simultaneously to reveal the essence of dynamics in IRES: how spatial variation in community structure changes through time in response to variation in the hydrological regimes of IRES over multiple spatial and temporal dimensions.

Modeling CPOM dynamics in IRES is a conceptually challenging task due to the strong influence of fragmented spatial mosaics and temporal hydrological pulses. Incorporating IRES into global carbon models and flux estimates will refine our understanding of the role played by river networks in global biogeochemical cycles and draw attention to the significant role probably played by IRES in these cycles.

More generally, it is essential to test whether *water* intermittence and *flow* intermittence produce similar relationships across different scale-dependent components of biodiversity (alpha, beta, gamma). Are the effects of flow intermittence across multiple spatial and temporal scales similar in IRES with and without hyporheic zones? Answer to these questions are important for management of flow regimes and restoration plans because they would inform guidelines for water policy decisions if the specific roles of water are identified as driving changes in biodiversity. Additionally, identifying how trends in biodiversity across different spatial components are influenced by flow intermittence in IRES would emphasize the value of biodiversity in IRES. This will also contribute to the knowledge of global biodiversity change, possibly significantly modifying general trends currently reported in the literature (e.g., McGill et al., 2015).

4.9.7 CONCLUSIONS

IRES are complex and dynamic systems. The structure and dynamics of metapopulations, metacommunities, and metaecosystems are inherently related to flow intermittence and to the associated patterns of hydrological connectivity (Chapter 2.3). The study of metapopulations in IRES is challenging due to the varying habitat template and the multiple dispersal pathways of organisms; accordingly, it requires modifying existing conceptual models and empirical approaches. Metacommunity studies have revealed that alpha, beta, and gamma diversities may be predictably associated with the degree of flow intermittence. Although the relevance of dispersal processes (and thus nested metacommunity structures) may be paramount during high-flow phases, species sorting (and thus metacommunity structures dominated by turnover) may be more important during nonflowing or dry phases. These ideas deserve further testing and serve as a suitable starting point to better understand the drivers of beta diversity in IRES. Finally, metaecosystem research in IRES has mainly focused on carbon dynamics, whereby the processed, transported, and stored portions of CPOM vary with changes in flow intermittence. There is now a need for further studies on other biogeochemical fluxes and their interactions with intermittence in different types of IRES.

Metasystem concepts are also very relevant for the management and conservation of IRES because this approach recognizes that sites are not isolated but connected by individuals, materials, and energy. If among-site dynamics are not well understood and explicitly considered, we may not be able to correctly appraise the resilience of these ecosystems, hindering optimized conservation strategies for species and communities. The intensification of the global hydrological cycle, combined with increasing exploitation of freshwater resources by humans, is expanding the extent and duration of hydrological droughts across many parts of the globe. Thus, we need to predict how alterations in patterns of flow intermittence induced by these global changes will affect riverine metapopulations, metacommunities, and metaecosystems if we are to successfully adapt management and conservation strategies for IRES (Chapters 5.4 and 5.5).

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